

Defect Log and Quality Metrics

****Course:**** Software Testing and Validation
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1. Defect Log

Defect ID	Description	Steps to Reproduce	Expected Result	Actual Result
Severity	Priority	Status		
DEF-001	30-day duration throws error	1. Book vehicle for exactly 30 days.	Booking succeeds (Total price calculated)	Validation error thrown: "Maximum rental period is 30 days."
DEF-002	Missing 'Premium Member' checkbox on form	1. Navigate to booking form.	Checkbox for Premium is visible.	Checkbox is missing from HTML template
DEF-003	Django ORM Async error in Playwright	1. Run pytest test_system.py	Tests pass	SynchronousOnlyOperation error thrown
DEF-004	Typos in vehicle list template	1. View vehicle list.	Headers spelled correctly	"Mkae" instead of "Make"

Defect Lifecycle Evidence

All defects followed the same lifecycle: ****New -> Assigned -> In Progress -> Retest -> Closed****.

Defect ID	Lifecycle evidence
DEF-001	New during boundary testing; assigned to the business-logic owner; fixed; retested with `TC_Dur_03`; closed.
DEF-002	New during UI inspection; assigned to the template owner; fixed; retested in the booking journey; closed.
DEF-003	New during system-test execution; assigned to the test/integration owner; fixed; Playwright suite rerun; closed.
DEF-004	New during UI inspection; assigned to the template owner; corrected; view test rerun; closed.

2. Quality Metrics

Metric	Calculation / Formula	Result	Interpretation
Defect Density	(Total Defects Found) / (Size of Application - e.g. KLOC or Function Points). Assuming ~0.5 KLOC. (4 / 0.5)	**8 defects / KLOC**	The codebase is very small, so the density appears artificially high, but 4 total defects is well within acceptable limits for an initial prototype.
Defect Removal Efficiency (DRE)	(Defects found pre-release) / (Total defects found pre-release + escaped defects) * 100. Assuming 0 escaped defects currently. (4 / (4+0)) * 100	**100%**	All known defects were caught by our automated test pyramid and fixed before code was merged to the main branch.
Code Coverage	Measured with `pytest --cov=rental --cov-branch`. (94% total; 100% branch coverage for `rental/business_logic.py`)	**94% total; 100% branch coverage**	The core business logic exceeds the 80% branch target. The remaining uncovered lines are mainly the seed command and defensive model/view paths.